

EDUCATION

Kochi, Kerala **Cochin University of Science and Technology** **Oct 2022 – May 2026**
Bachelor of Technology in Computer Science & Engineering- **4th Year (7th Semester)**

Academic Achievements:

- CGPA: **9.6/10**
- Represented **India** at **European Rover Challenge 2024**, Poland (Team: Horizon, World Rank: 18th)
- Listed among **Top 10% Students** in the Batch.

PROFESSIONAL EXPERIENCE

Software Developer Intern **Horizon Mars Rover, CUSAT** **Nov 2023 - Feb 2024**

- Developed and implemented a URDF-based robotic arm model with TF and Gazebo in ROS 2, including forward and inverse kinematics.
- Controlled a robotic arm using Arduino Due (32-bit ARM) and servo motors, improving motion precision by **35%** through real-time kinematics integration.
- **Technologies used : C++, ROS 2 (Humble), Linux, Arduino IDE**

Embedded Intern **Brainspired Labs Pvt. Ltd.** **Jan 2025 - June 2025**

- Integrated dual 16-bit DAC with STM32H753ZI via I2C, improving signal accuracy by **40%**. Enabled UART communication on ESP32 with Android app for real-time frequency and voltage control.
- Integrated CJMCU-1334 audio DAC to esp32-s3 using I2S protocol & gained hands-on experience with AD5750 eval board.
- **Technologies used : STM32CUBE-IDE, Arduino IDE, Embedded C**

POSITION OF RESPONSIBILITIES

SIG AI Lead **IEEE Execom'24** **Jun 2024 – Dec 2024**

- Co-managed 5+ AI-related tech events at college in collaboration with other IEEE Leads.
- Conducted training sessions for juniors on AI and ML with 300-400 attendees.
- Sharpened my communication skill and learned to work in a team environment.

Team Lead **Horizon Mars Rover, CUSAT** **Oct 2024 – Present**

- Technical team of college responsible for building semi-autonomous Mars rover for competition.
- Led Mars Rover development by setting goals and managing cross-functional teams.
- Ensured effective collaboration across mechanical, electronics, and software divisions (also worked as Full stack engineer).

PERSONAL PROJECTS

- **Diksha(2024)🔗**: Web app to learn & get educational tools for free. A **MERN Stack, Vertex AI & Gemini API** Project that leverages Problem Solving Tools, NEP Syllabus Generator, AI bot, Grading System & more.
- **Preventive AI(2024)🔗**: AI platform for early cancer detection and personalized treatment using imaging, genomics & EHRs. Built with **Python, ML, TensorFlow Serving, Node.js & MongoDB**. Processed structured and unstructured medical data in **simulated Big Data pipelines** with **87%** model accuracy.
- **Mobile Robot Simulation(2025)🔗**: Developed and simulated an autonomous mobile robot with a robotic arm in **ROS 2 using URDF, SLAM, Nav2, and Gazebo**.

ACHIEVEMENTS

- Won **5 Hackathons** (out of 10) conducted by **Google Development Group 🔗 (GDG)** , **Major League Hacking 🔗** and **Inovact 🔗**.
- Solved **100+** programming problems across various platforms like LeetCode, Hackerrank etc.
- **Jury Choice Award Winner** at AIC by **Accenture & Finalist** at DevsHouse'24 Hackathon by VIT Chennai.

SKILLS

Languages: Java(Advanced), JavaScript, TypeScript, HTML/CSS, C/C++, SQL, Python, NoSQL, OOPs

Technical Skills: Git & GitHub, DSA, SDLC, AWS, Node.js, Express.js, Linux, Unit Testing, Agile, RESTful APIs, GraphQL, Kafka, NLP, AI-ML, Kubernetes, UI/UX, Embedded Systems (STM32, ESP3, DACs).